



Lab Task 2

Only for course Teacher					
	Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage	25%	50%	75%	100%	
Clarity					
Content Quality					
Spelling & Grammar					
Organization and Formatting					
Total obtained mark					
Comments					

Semester: Spring / Fall

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Batch: 46

Section:B

Course Code: CSE 124

Course Name: Object Oriented Programming

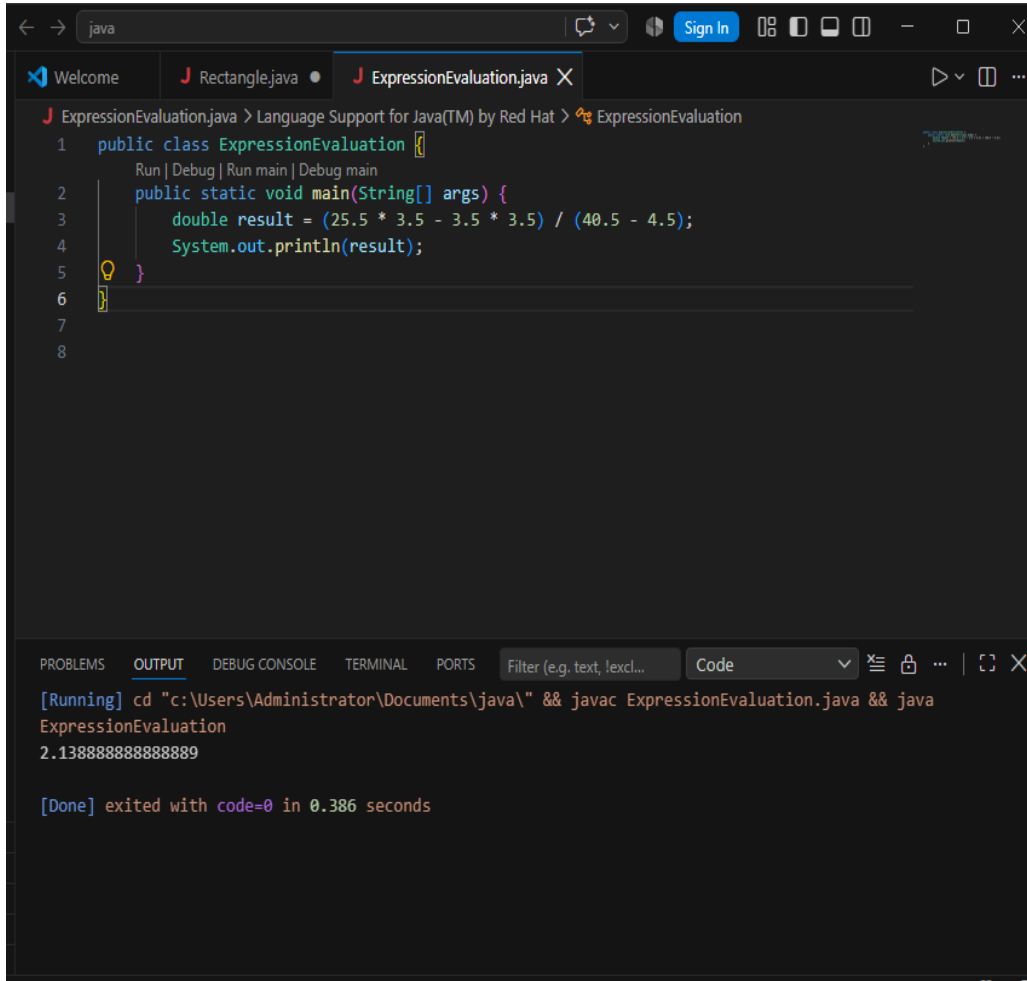
Course Teacher Name: Debabarata Mallick

Designation:Lecturer

Submission Date :13\ 08\ 2026

Experiment No : 09

Experiment Name : Expression Evaluation .



The screenshot shows an IDE window with a Java file named `ExpressionEvaluation.java`. The code defines a public class `ExpressionEvaluation` with a `main` method. Inside the `main` method, a `double` variable `result` is calculated using the expression $(25.5 * 3.5 - 3.5 * 3.5) / (40.5 - 4.5)$, and the result is printed to the console. Below the code editor, the `OUTPUT` tab is selected, showing the execution of the program. The output displays the command used to compile and run the program, followed by the calculated result `2.138888888888889` and a confirmation that the program exited successfully.

```
1 public class ExpressionEvaluation {  
    Run | Debug | Run main | Debug main  
2     public static void main(String[] args) {  
3         double result = (25.5 * 3.5 - 3.5 * 3.5) / (40.5 - 4.5);  
4         System.out.println(result);  
5     }  
6 }  
7  
8
```

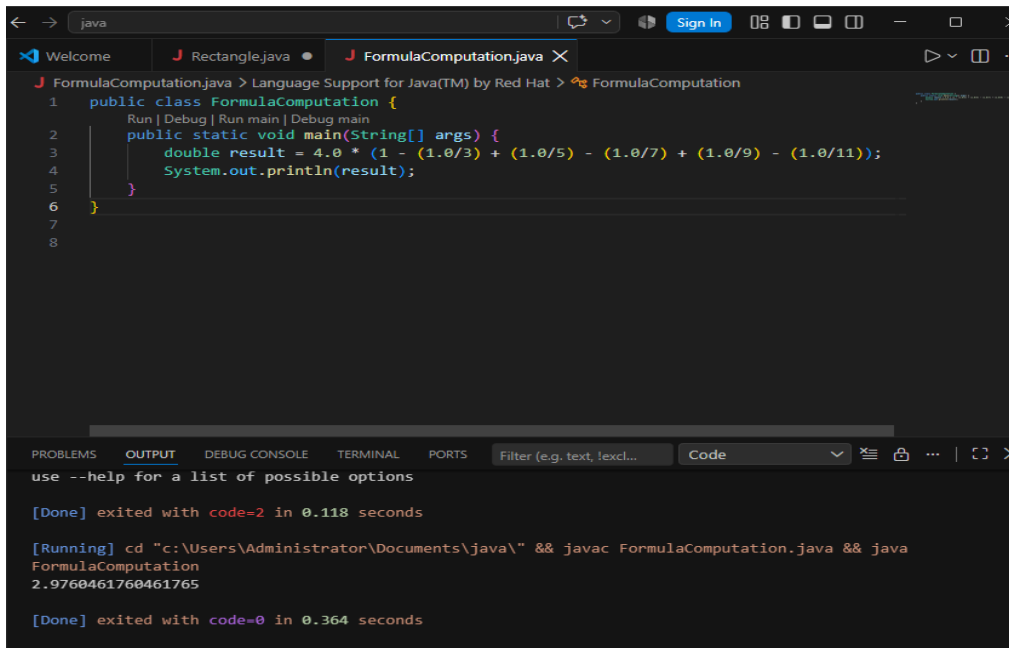
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, lexcl... Code

[Running] cd "c:\Users\Administrator\Documents\java\" && javac ExpressionEvaluation.java && java ExpressionEvaluation
2.138888888888889
[Done] exited with code=0 in 0.386 seconds

Discussion: This program calculates a mathematical expression with multiplication and division. I followed the given formula step by step. The final result came out to be approximately 2.14.

Experiment No : 10

Experiment Name : Formula Computation.



The screenshot shows an IDE window with two tabs: 'Rectangle.java' and 'FormulaComputation.java'. The 'FormulaComputation.java' tab is active, displaying the following Java code:

```
1 public class FormulaComputation {
2     Run | Debug | Run main | Debug main
3     public static void main(String[] args) {
4         double result = 4.0 * (1 - (1.0/3) + (1.0/5) - (1.0/7) + (1.0/9) - (1.0/11));
5         System.out.println(result);
6     }
7
8 }
```

Below the code editor, the 'OUTPUT' tab is selected, showing the execution results:

```
use --help for a list of possible options

[Done] exited with code=2 in 0.118 seconds

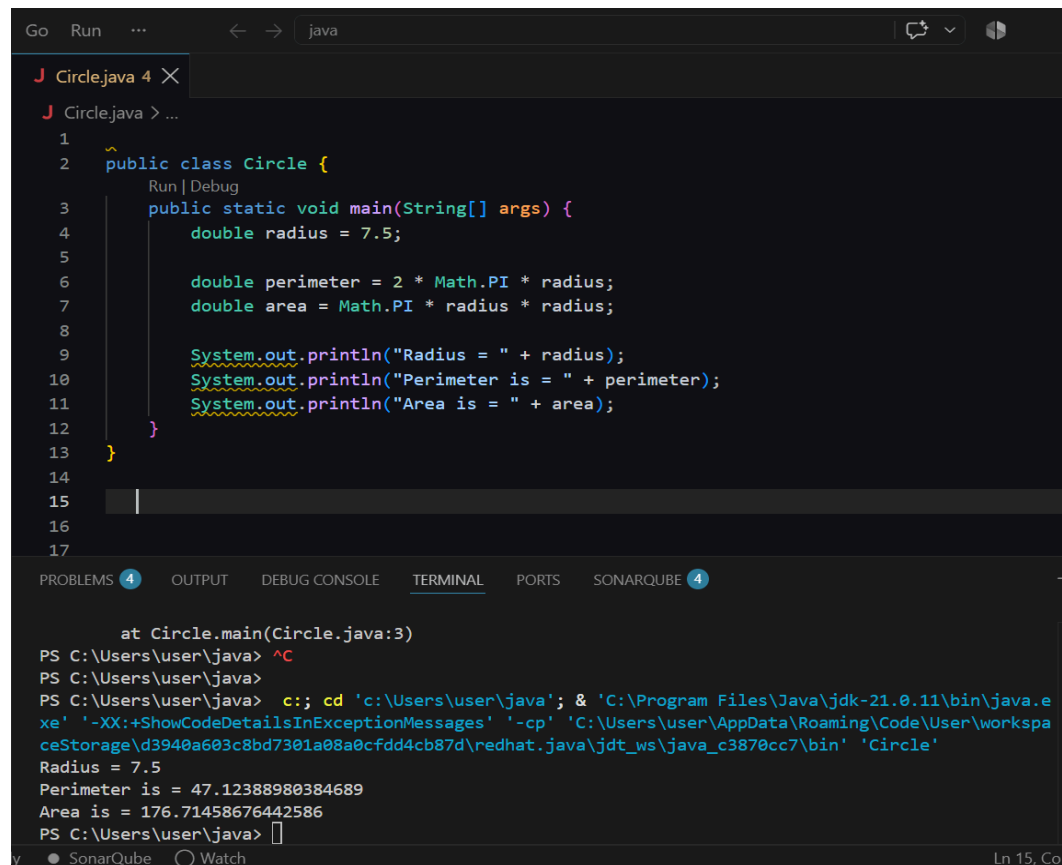
[Running] cd "c:\Users\Administrator\Documents\java\" && javac FormulaComputation.java && java
FormulaComputation
2.9760461760461765

[Done] exited with code=0 in 0.364 seconds
```

Discussion : This formula is used to find an approximate value of pi. I calculated the given terms using floating point numbers. The output I got is around 2.976.

Experiment No: 11

Experiment Name : Circle Area And Perimeter



The screenshot shows an IDE window with a file named 'Circle.java'. The code defines a 'Circle' class with a 'main' method. Inside 'main', a radius of 7.5 is assigned. The perimeter is calculated as $2 * \text{Math.PI} * \text{radius}$ and the area is calculated as $\text{Math.PI} * \text{radius} * \text{radius}$. Both results are printed to the console. The bottom panel shows the terminal output, which matches the code's output: 'Radius = 7.5', 'Perimeter is = 47.12388980384689', and 'Area is = 176.71458676442586'.

```
1 public class Circle {
2     public static void main(String[] args) {
3         double radius = 7.5;
4
5         double perimeter = 2 * Math.PI * radius;
6         double area = Math.PI * radius * radius;
7
8         System.out.println("Radius = " + radius);
9         System.out.println("Perimeter is = " + perimeter);
10        System.out.println("Area is = " + area);
11    }
12 }
13
14
15
16
17
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS SONARQUBE 4

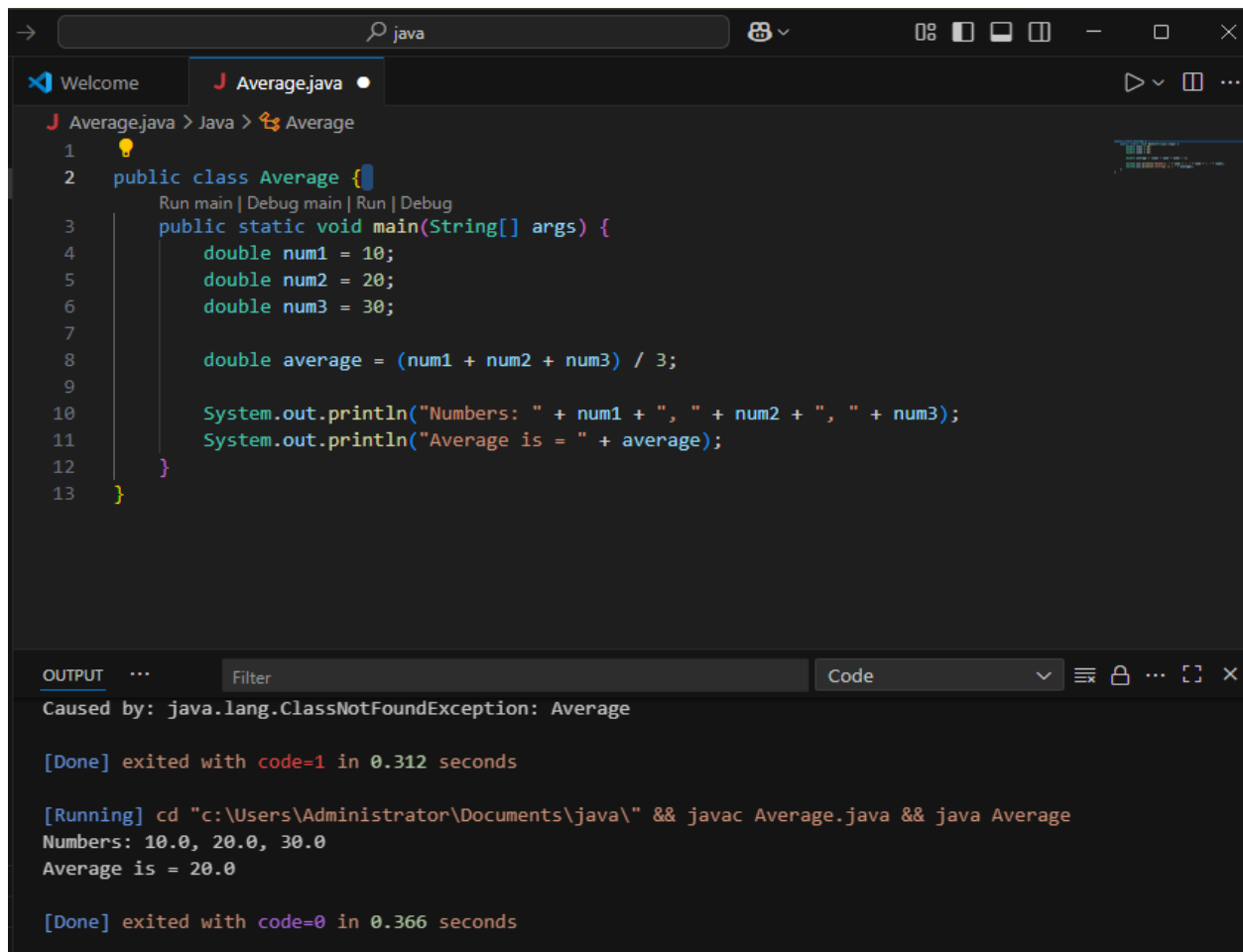
at Circle.main(Circle.java:3)
PS C:\Users\user\java> ^C
PS C:\Users\user\java>
PS C:\Users\user\java> c:: cd 'c:\Users\user\java'; & 'C:\Program Files\Java\jdk-21.0.11\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\user\AppData\Roaming\Code\User\workspaceStorage\d3940a603c8bd7301a08a0cfdd4cb87d\redhat.java\jdt_ws\java_c3870cc7\bin' 'Circle'
Radius = 7.5
Perimeter is = 47.12388980384689
Area is = 176.71458676442586
PS C:\Users\user\java> |

Ln 15, Col 1

Discussion: I calculated the area and perimeter of a circle using radius 7.5. Used the formulas $2\pi r$ for perimeter and πr^2 for area.

Experiment No: 12.

Experiment Name : Average Of Three Numbers



The screenshot shows an IDE window with a file named `Average.java`. The code defines a public class `Average` with a `main` method. Inside `main`, three double variables `num1`, `num2`, and `num3` are initialized with values 10, 20, and 30 respectively. The average is calculated as $(num1 + num2 + num3) / 3$. The program then prints the numbers and the calculated average.

```
1 public class Average {  
2     public static void main(String[] args) {  
3         double num1 = 10;  
4         double num2 = 20;  
5         double num3 = 30;  
6  
7         double average = (num1 + num2 + num3) / 3;  
8  
9         System.out.println("Numbers: " + num1 + ", " + num2 + ", " + num3);  
10        System.out.println("Average is = " + average);  
11    }  
12 }  
13 }
```

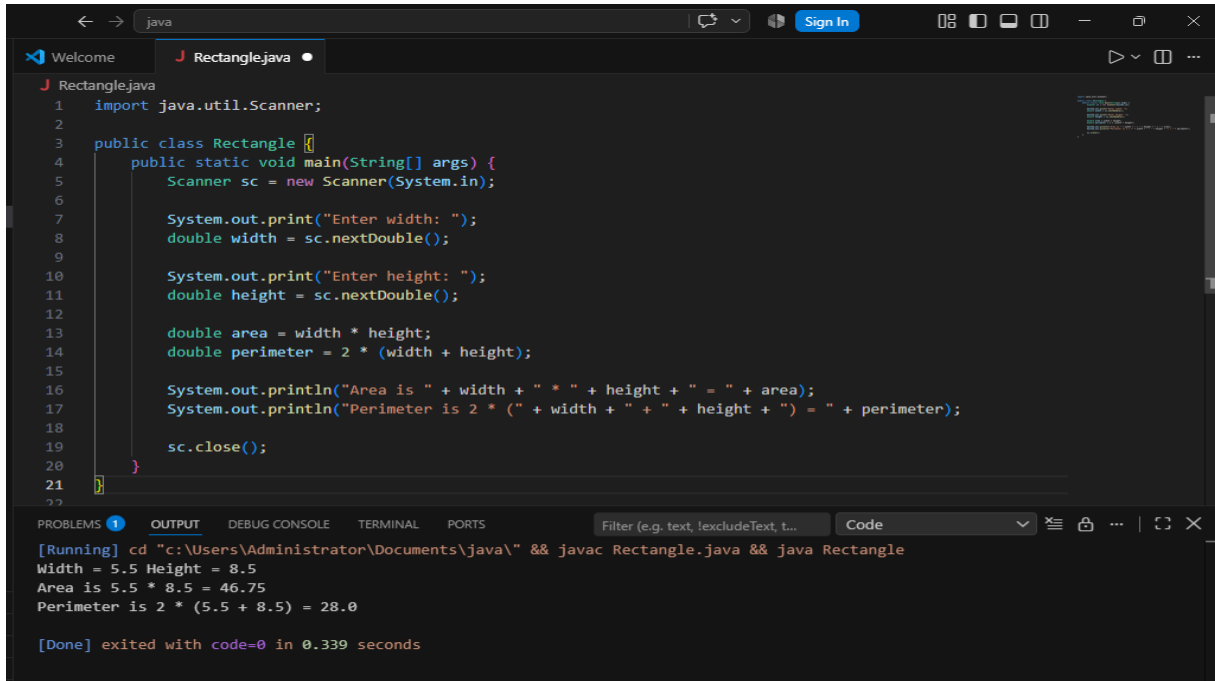
The output window shows the following text:

```
Caused by: java.lang.ClassNotFoundException: Average  
  
[Done] exited with code=1 in 0.312 seconds  
  
[Running] cd "c:\Users\Administrator\Documents\java\" && javac Average.java && java Average  
Numbers: 10.0, 20.0, 30.0  
Average is = 20.0  
  
[Done] exited with code=0 in 0.366 seconds
```

Discussion: I took three numbers and found their average by adding them and dividing by 3 .

Experiment No: 13.

Experiment Name : Rectangle Area And Perimeter .



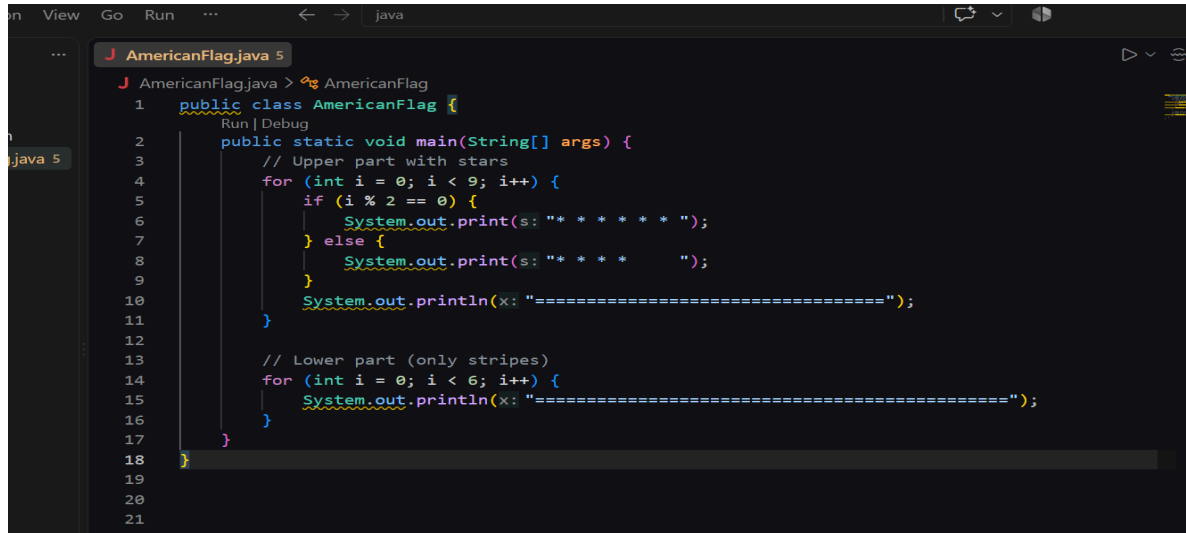
```
1 import java.util.Scanner;
2
3 public class Rectangle {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter width: ");
8         double width = sc.nextDouble();
9
10        System.out.print("Enter height: ");
11        double height = sc.nextDouble();
12
13        double area = width * height;
14        double perimeter = 2 * (width + height);
15
16        System.out.println("Area is " + width + " * " + height + " = " + area);
17        System.out.println("Perimeter is 2 * (" + width + " + " + height + ") = " + perimeter);
18
19        sc.close();
20    }
21 }
```

[Running] cd "c:\Users\Administrator\Documents\java\" && javac Rectangle.java && java Rectangle
Width = 5.5 Height = 8.5
Area is 5.5 * 8.5 = 46.75
Perimeter is 2 * (5.5 + 8.5) = 28.0
[Done] exited with code=0 in 0.339 seconds

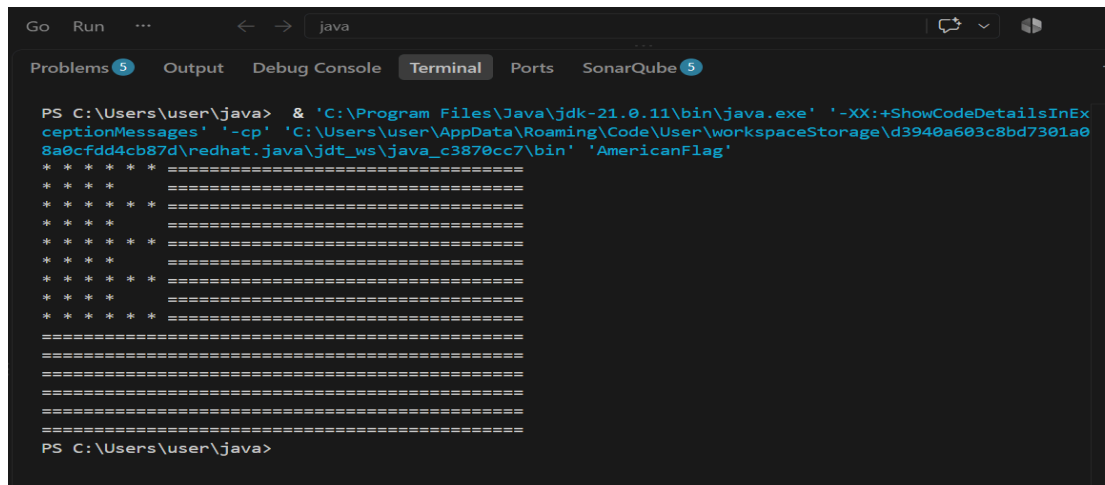
Discussion : I calculate the area and perimeter of a rectangle using width 5.5 and height 8.5 with the formulas length and width and $2(l+w)$.

Experiment No : 14

Experiment Name : American Flag Display .



```
1 public class AmericanFlag {
2     public static void main(String[] args) {
3         // Upper part with stars
4         for (int i = 0; i < 9; i++) {
5             if (i % 2 == 0) {
6                 System.out.print(s: "* * * * * ");
7             } else {
8                 System.out.print(s: "* * * * ");
9             }
10            System.out.println(x: "=====");
11        }
12
13        // Lower part (only stripes)
14        for (int i = 0; i < 6; i++) {
15            System.out.println(x: "=====");
16        }
17    }
18 }
19
20
21
```



```
PS C:\Users\user\java> & 'C:\Program Files\Java\jdk-21.0.11\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\user\AppData\Roaming\Code\User\workspaceStorage\d3940a603c8bd7301a08a0cfd4cb87d\redhat.java\jdt_ws\java_c3870cc7\bin' 'AmericanFlag'
* * * * * =====
* * * * * =====
* * * * * =====
* * * * * =====
* * * * * =====
* * * * * =====
* * * * * =====
* * * * * =====
* * * * * =====
=====
=====
=====
=====
=====
=====
PS C:\Users\user\java>
```

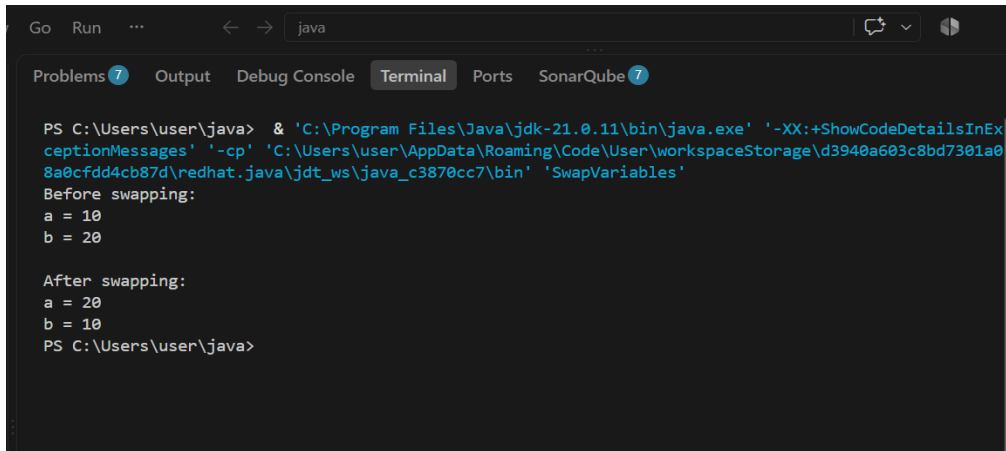
Discussion:I printed an american flag pattern using loops with stars and equal signs to make the stars.

Experiment No : 15

Experiment Name: Swap Two Variables.



```
Go Run ... < -> java
J SwapVariables.java 7
J SwapVariables.java > SwapVariables
1 public class SwapVariables {
    Run | Debug
2     public static void main(String[] args) {
3         int a = 10;
4         int b = 20;
5
6         System.out.println(x: "Before swapping:");
7         System.out.println("a = " + a);
8         System.out.println("b = " + b);
9
10        // Swapping using a temporary variable
11        int temp = a;
12        a = b;
13        b = temp;
14
15        System.out.println(x: "\nAfter swapping:");
16        System.out.println("a = " + a);
17        System.out.println("b = " + b);
18    }
19 }
```



```
Go Run ... < -> java
Problems 7 Output Debug Console Terminal Ports SonarQube 7
PS C:\Users\user\java> & 'C:\Program Files\Java\jdk-21.0.11\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\user\AppData\Roaming\Code\User\workspaceStorage\d3940a603c8bd7301a08a0cfdd4cb87d\redhat.java\jdt_ws\java_c3870cc7\bin' 'SwapVariables'
Before swapping:
a = 10
b = 20

After swapping:
a = 20
b = 10
PS C:\Users\user\java>
```

Discussion : I swapped the values of two variables using a temporary variable and printed the values before and after swapping.